# Forecasting USDZAR one month ahead

## a forecast combination against the AR(1) benchmark and the random walk

SES x Codera Analytics Hackathon, July 2026

Team: Spurious Decision
Aakash Anil and Stefan von der Heyden
Stellenbosch University

Built from the saved pipeline outputs.

# 1. Summary

We forecast USDZAR four weeks ahead at 261 Fridays, from 1 January 2021 to 26 December 2025, and scored the forecasts against the mandated AR(1) benchmark. Our forecast combination, an average of a random walk, a carry forecast and a shrunk ridge regression, scored an RMSE of 0.5599 against 0.5624 for the AR(1), which is 0.44 per cent lower. The difference is not statistically established. The Diebold-Mariano p-value is 0.449, and the 95% interval for the mean squared-error difference runs from -0.0099 to +0.0044, which includes zero. Against the random walk, whose RMSE is 0.5600, the combination is level (p = 0.967). In practice it is the random walk with a small tilt: its forecasts sit 0.089 per cent from spot on average.

Two further results limit what the combination can claim. An equal-weight average of the same three components scored 0.5599, so the adaptive weighting contributes nothing we can measure. The weights stayed between 0.306 and 0.353, because the components' errors are almost the same series (correlation up to 0.99998). And the one flexible model we ran, gradient boosting, scored 0.5745, the highest of the 13 models. That conclusion covers this specification and nothing broader.

The AR(1) differs from the other models in a way we can measure. Its implied long-run level sat below spot at 261 of 261 origins, by R2.11 on average, and it never forecast a rise. The difference in squared bias accounts for 55 per cent of its excess squared error against the combination. We report this as an observation and have not shown that it is the whole explanation.

All 2021 to 2025 results are exploratory. The window was scored repeatedly while the code was built and repaired, comparisons were added after earlier results had been seen, and no new holdout exists. Section 7 sets out a protocol for a prospective test.

# 2. Task and protocol

The brief is to forecast USDZAR (SARB series EXCX135, distributed through EconData) one month ahead, every Friday, out of sample, over 2021 to 2025, and to compare RMSE against an AR(1) model.

We read it this way. Every Friday t from 1 January 2021 to 26 December 2025 is an origin, which gives 261 of them. Spot at t is the last EXCX135 fixing dated on or before t. The target is the EXCX135 fixing 28 calendar days later, or the last fixing before that date when it falls on a holiday. The last target falls on 23 January 2026, after the end of 2025, and 257 of the 261 targets fall inside 2025. Twenty-eight days keeps origin and target on the same weekday. We did not rerun the exercise with calendar months. The information set at t holds only values that were public by 17:00 Johannesburg time on the Friday, each joined on its publication time, with the vintage then current for series that are revised (section 3).

# 3. Data

**Target series.** EXCX135 is the rand per US dollar rate, calculated as the weighted average of the banks' daily rates at approximately 10:30 and weighted by their foreign exchange turnover. The SARB publishes it and EconData distributes it. We use the official EconData export as the primary source: dataset MARKET_RATES v1.0.0, series key EXCX135.B.A, 14,173 daily observations from 2 January 1970. The pipeline also downloads the same series from the SARB Web API under code EXCX135D, and verify_provenance.py compares the two: 9,141 overlapping observations, largest absolute difference 0.000000, 0 mismatches, including all 1,290 observations in the forecast window. Citation: SARB. MARKET_RATES. Version 1.0.0. Pretoria: SARB [producer]. Johannesburg: EconData [distributor], 2026.

**Predictors.** JIBAR 3M from the SARB and the US 3M Treasury bill from FRED, whose difference is the carry signal. Gold at the London fixing, platinum futures from NYMEX and Brent from FRED make up the commodity block: South Africa exports two of those and imports the third. The Fed's broad dollar index and the VIX capture dollar strength and global risk appetite. Technical variables complete the set: the rand's own four- and twelve-week returns and its gap to the 200-day moving average. Estimation history starts in 1999.

**When each value is known.** The forecast is issued at 17:00 Johannesburg time. The SARB rate and JIBAR are public that morning. The gold fix is public that afternoon in UK summer and a day earlier in UK winter. Platinum and the VIX are available to the Thursday close, the T-bill rate to Wednesday, Brent to the Monday and the dollar index to the previous Friday. The dollar index, Brent, the T-bill rate and the VIX are revised, so each takes the value ALFRED, the St. Louis Fed's vintage archive, showed at the time, and a four-week return takes both ends from one vintage. Before a series' first vintage (for the dollar index, 4 February 2019) the earliest vintage stands in, which is an approximation. Each series' rule, source and reasoning is in data/raw/series_metadata.json. The gold rule is the least certain: the 30 minutes after the auction is our allowance, not a published time. Section 7 reports a rerun with every non-spot predictor a further week late.

**What is left out.** We used no inflation, current-account or GDP series. They are published late and revised, and we hold no vintages for them. Using today's published history in a simulated 2021 forecast would use numbers that did not exist then.

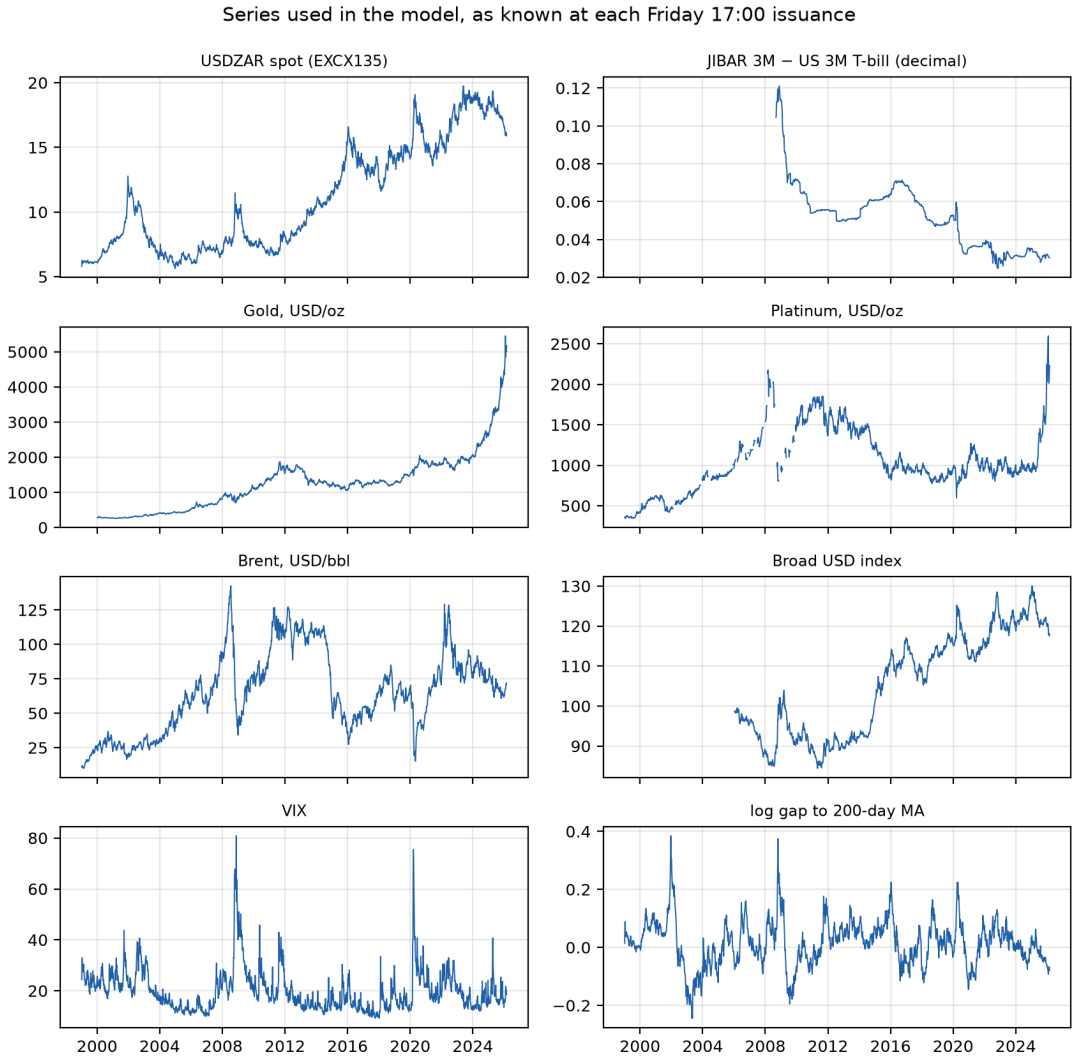



Figure 3.1. Every series used in the models, as known at each Friday 17:00 issuance.

Summary statistics - weekly (Friday) series

| | obs before 2021 | obs 2021-25 | mean | std | min | max |
|---|---|---|---|---|---|---|
| spot | 1147.0 | 261.0 | 11.042 | 4.17 | 5.636 | 19.745 |
| carry_diff | 643.0 | 261.0 | 0.052 | 0.017 | 0.025 | 0.121 |
| gold | 1095.0 | 261.0 | 1248.135 | 789.295 | 258.375 | 5453.35 |
| platinum | 1046.0 | 261.0 | 1047.028 | 387.574 | 344.1 | 2597.4 |
| brent | 1147.0 | 261.0 | 65.031 | 29.258 | 9.97 | 142.43 |
| usd_broad | 781.0 | 261.0 | 105.311 | 12.835 | 84.424 | 130.051 |
| vix | 1147.0 | 261.0 | 19.969 | 8.267 | 9.19 | 80.86 |

Table 3.1. Observation counts split between the estimation era before 2021 and the forecast era from 2021 to 2025, with descriptive statistics.

# 4. Methodology

## 4.1 The baseline

AR(1) with intercept on the weekly log USDZAR level, re-estimated by OLS on an expanding window at every origin and iterated four steps ahead. Its judged-window RMSE is 0.5624. It was fixed as the benchmark before scoring. Three variants appear only as exploratory comparisons: the same model on the level (0.5629), on a five-year window (0.5724), and with a retransformation correction (0.5617). The correction matters because the exponential of a forecast of the log is a median, not a mean, under normal errors. For the log AR(1) the level the forecast is pulled towards is exp(c/(1-phi)), the equilibrium of the log. It is fragile near a unit root: phi ran from 0.9971 to 0.9986 across the origins, so 1-phi ran from 0.0014 to 0.0029, and the level ranged from R11.80 to R19.18. At the worked-example origin, 3 June 2022, the fit on 1,222 weekly observations gave c = 0.006335 and phi = 0.997531, a level of R13.01 and a shock half-life of 280 weeks. Spot that day was R15.41.

## 4.2 Component models

**Random walk.** The forecast is today's rate. **Carry.** Covered interest parity ties the forward rate to the interest differential, so the forecast is the spot rate multiplied by one plus the JIBAR minus T-bill gap scaled to 28 days. A forward prices a contract that can be locked in today. It is not the expected future spot rate, and uncovered parity generally fails, so we treat carry as one component and make no claim that it forecasts well alone. The inputs also differ in kind (a bank rate against a government bill) and in tenor (three months against a 28-day horizon). **Ridge regression.** The four-week log return regressed on the the full predictor set, no intercept: carry differential and its four-week change, four-week returns on gold, platinum, Brent, the broad dollar and the rand, the twelve-week rand return, the VIX z-score and its four-week change, and the gap to the moving average. The penalty is re-selected every 13 origins by five-fold time-ordered cross-validation. A training row whose target date falls on or after the start of the fold it is scored against is purged, and the predictors are standardised inside each fold. Training rows are restricted to origins whose 28-day outcome had already resolved. With no intercept, heavy shrinkage collapses the model to the random walk. **Gradient boosting.** A HistGradientBoosting comparator on the same rows. Every 52 weeks the same purged cross-validation picks its depth, learning rate, iteration count, L2 penalty and leaf size from 12 configurations.

## 4.3 The combination and the selection step

The combination weights the random walk, carry and ridge by inverse discounted mean squared error over resolved past forecasts only, with a 28-day embargo, a window of 156 resolved origins, decay 0.97 and sharpness 2. The ridge variant, the decay and the sharpness were selected on 309 validation origins from 2 January 2015 to 27 November 2020. The last of them is the final Friday whose target, 25 December 2020, falls before 2021, and the code asserts it. The chosen ridge variant is RIDGE_f0 and the full grid of 24 configurations is in validation_metrics.json. Validation scores were used to pick the configuration, so they are not independent evidence. The surface is flat: the fifth-best configuration is 0.00012 worse than the best. Weights for the first judged forecasts were warmed up on walk-forward forecasts from 2013.

# 5. Results

| Model | RMSE | MAE | RMSE below AR(1) | Rank | Calls a rise |
|---|---|---|---|---|---|
| AR(1), log level (benchmark) | 0.5624 | 0.4478 | | 9 | 0% |
| AR(1), level | 0.5629 | 0.4492 | -0.09% | 11 | 31% |
| AR(1), 5-year window | 0.5724 | 0.4502 | -1.79% | 12 | 9% |
| AR(1), mean-corrected | 0.5617 | 0.4492 | +0.11% | 8 | 59% |
| Random walk | 0.5600 | 0.4480 | +0.43% | 4 | 0% |
| Random walk with drift | 0.5627 | 0.4577 | -0.05% | 10 | 100% |
| Carry | 0.5606 | 0.4540 | +0.32% | 7 | 100% |
| Ridge | 0.5602 | 0.4479 | +0.38% | 6 | 72% |
| Gradient boosting | 0.5745 | 0.4723 | -2.16% | 13 | 65% |
| **Combination** | **0.5599** | **0.4495** | **+0.44%** | **3** | **100%** |
| Equal-weight RW + carry + ridge | 0.5599 | 0.4495 | +0.44% | 2 | 100% |
| Adaptive RW + carry | 0.5599 | 0.4504 | +0.44% | 1 | 100% |
| Adaptive RW + ridge | 0.5601 | 0.4479 | +0.40% | 5 | 72% |

Table 5.1. Judged window, n = 261. RMSE below AR(1) is the per cent by which a model's RMSE is under the benchmark's, so a positive value favours the model. Rank is by RMSE across the 13 models. The first row is the mandated benchmark; the AR(1) variants, the drift, the equal-weight average and the two adaptive pairs are exploratory. On the strict cut, with targets inside 2025 only, the combination scores 0.5619, the AR(1) 0.5647 and the random walk 0.5622. That cut is a subset of the same data.

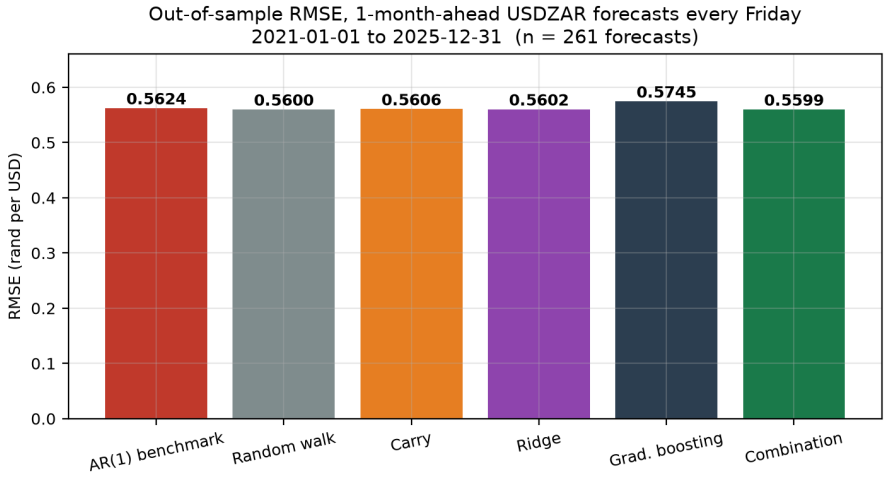



Figure 5.1. RMSE by model. The axis starts at zero, so the bars look level: the differences are in the third decimal place. Figure 5.2 shows them with intervals.

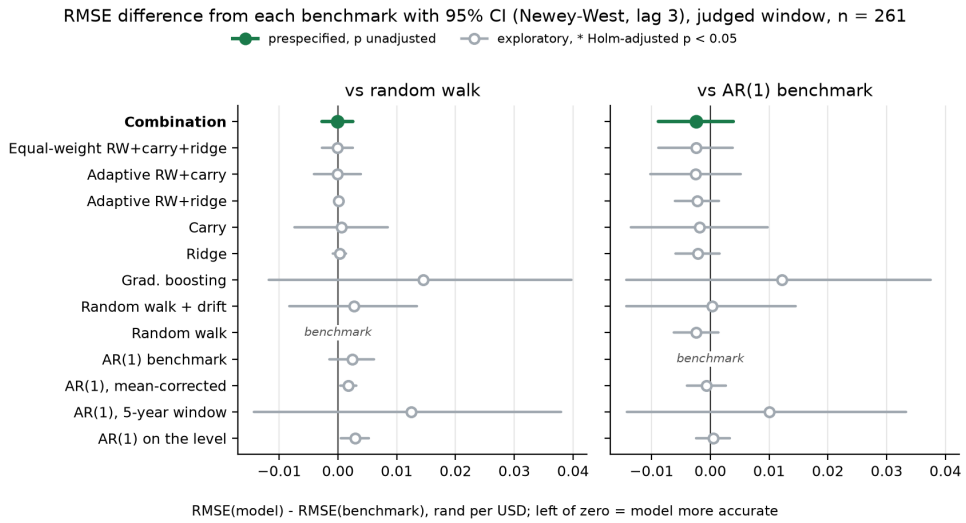



Figure 5.2. RMSE difference from the random walk and from the AR(1), with 95% Newey-West intervals (lag 3). Filled points are the two prespecified comparisons. Left of zero means the model is more accurate.

| Comparison | DM (lag 3) | p | 95% interval, mean squared-error difference | RMSE difference | p across four lags |
|---|---|---|---|---|---|
| Combination vs AR(1) | -0.76 | 0.449 | -0.0099 to +0.0044 | -0.00245 | 0.449 to 0.456 |
| Combination vs random walk | -0.04 | 0.967 | -0.0030 to +0.0029 | -0.00006 | 0.966 to 0.967 |

Table 5.2. The two prespecified comparisons. DM is the Diebold-Mariano statistic on squared error with a Newey-West variance; a negative value favours the combination. The last column shows the range of p-values over HAC lags 3, 4, 8 and the Andrews (1991) automatic choice, which was 13 against the AR(1) and 11 against the random walk. The RMSE interval against the AR(1), holding the benchmark's MSE fixed, is -0.0088 to +0.0039.

| Model | DM vs AR(1) | p | Holm p | DM vs RW | p | Holm p |
|---|---|---|---|---|---|---|
| AR(1), log level (benchmark) | - | - | - | +1.23 | 0.218 | 1.000 |
| AR(1), level | +0.33 | 0.741 | 1.000 | +2.39 | 0.017 | 0.338 |
| AR(1), 5-year window | +0.82 | 0.410 | 1.000 | +0.92 | 0.355 | 1.000 |
| AR(1), mean-corrected | -0.38 | 0.703 | 1.000 | +2.59 | 0.010 | 0.201 |
| Random walk | -1.23 | 0.218 | 1.000 | - | - | - |
| Random walk with drift | +0.04 | 0.968 | 1.000 | +0.48 | 0.628 | 1.000 |
| Carry | -0.31 | 0.760 | 1.000 | +0.14 | 0.885 | 1.000 |
| Ridge | -1.12 | 0.264 | 1.000 | +0.42 | 0.673 | 1.000 |
| Gradient boosting | +0.91 | 0.363 | 1.000 | +1.10 | 0.273 | 1.000 |
| **Combination (prespecified)** | **-0.76** | **0.449** | **n/a** | **-0.04** | **0.967** | **n/a** |
| Equal-weight RW + carry + ridge | -0.77 | 0.444 | 1.000 | -0.06 | 0.952 | 1.000 |
| Adaptive RW + carry | -0.63 | 0.525 | 1.000 | -0.05 | 0.962 | 1.000 |
| Adaptive RW + ridge | -1.19 | 0.233 | 1.000 | +0.39 | 0.697 | 1.000 |

Table 5.3. Every model against both benchmarks at HAC lag 3. The 21 distinct exploratory hypotheses are adjusted together by Holm's method (2 have an unadjusted p below 0.05, 0 survive). n/a marks the prespecified comparisons, which are not adjusted.

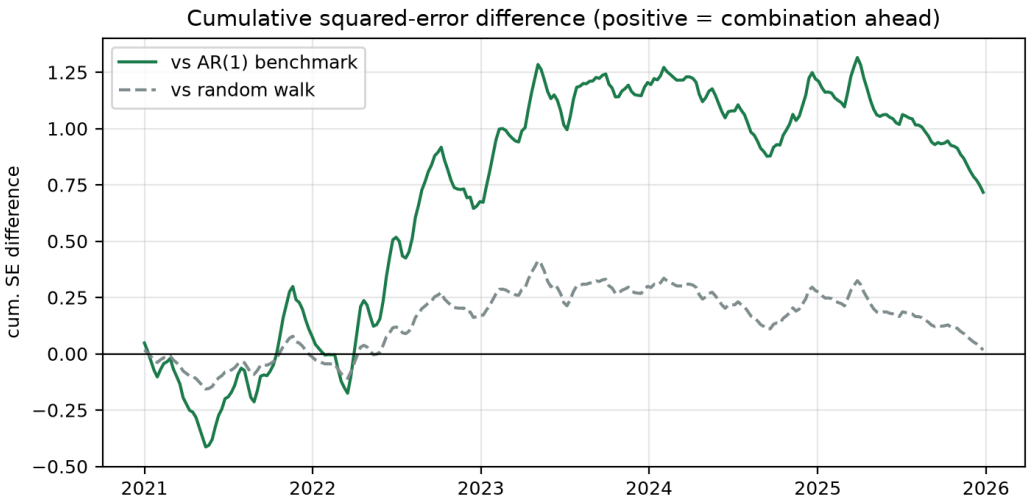



Figure 5.3. Cumulative squared-error difference, positive when the combination is ahead. Against the AR(1) it ends 2021 at 0.08, 2022 at 0.68, 2023 at 1.20 and 2024 at 1.22, peaks at 1.32 on 28 March 2025 and finishes at 0.72. The line against the random walk stays much closer to zero.

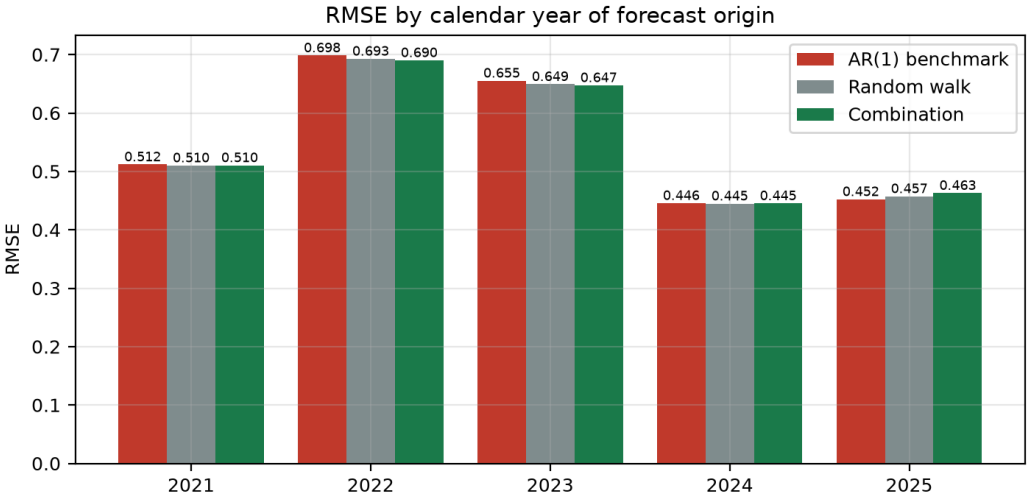



Figure 5.4. RMSE by calendar year of origin. The combination is lower than the AR(1) in 4 of 5 years (2021, 2022, 2023 and 2024). In 2025 the AR(1) scored 0.4523 against 0.4629 for the combination. We have not tested why. Five annual figures cannot show whether the pattern is stable.

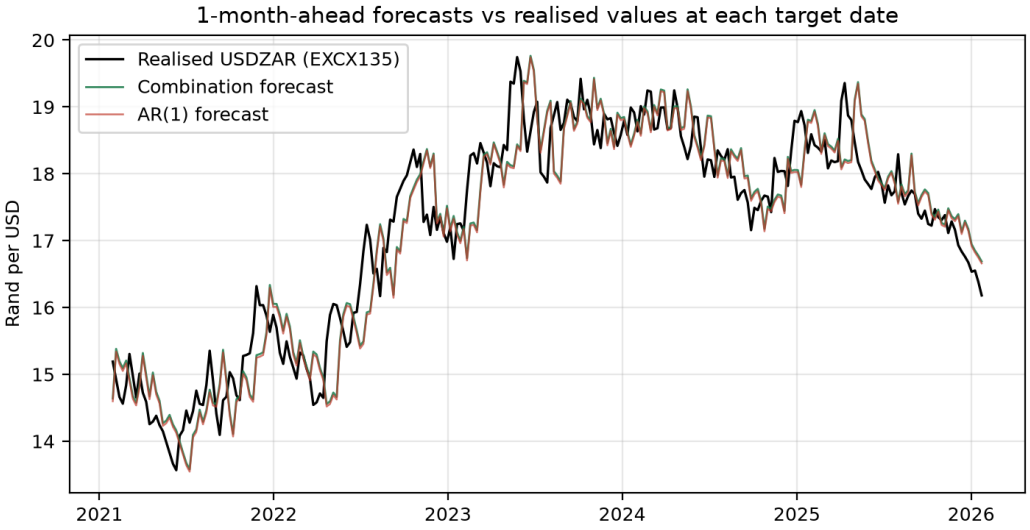



Figure 5.5. Forecasts plotted at their target dates against the realised series.

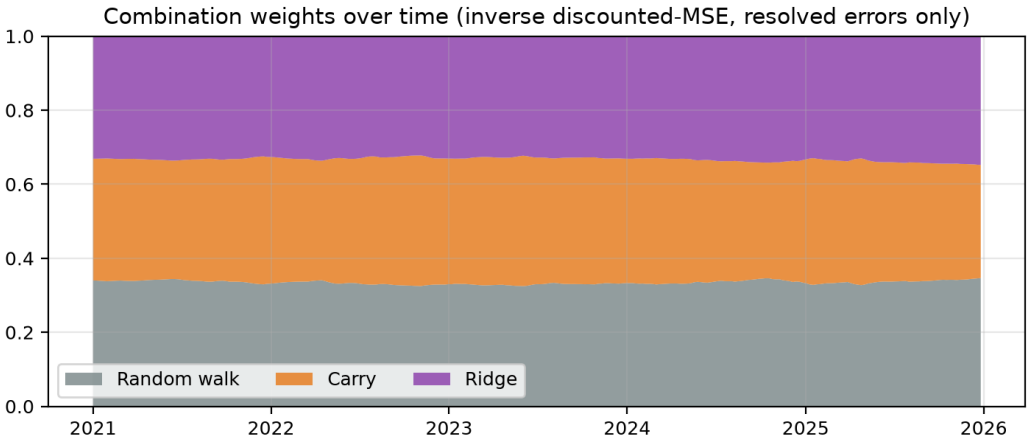



Figure 5.6. Combination weights through time, computed from resolved errors only. They stay between 0.306 and 0.353, close to one third each.

# 6. Discussion

The differences between the leading models are small: the 7 models built from the random walk, carry and ridge fall within 0.00068 of each other in RMSE. A common explanation is that USDZAR is priced by a deep market, so public information is in the price quickly and a month-ahead forecast is left with a small residual. That is plausible and consistent with Meese and Rogoff (1983) and the survey by Rossi (2013), but nothing here tests it.

The combination is almost the random walk. Its forecasts differ from spot by 0.089 per cent on average. It forecast a rise at 100 per cent of origins, because carry's tilt is always upward, while the rand rose at 46.4 per cent of them, so its directional accuracy (46.4 per cent) is just the share of rises. It carries no directional information. Adaptive weighting did not help: equal weights scored 0.5599 against 0.5599, and the largest gap among the three ablations is 0.00017.

What we can say about the AR(1) is descriptive. It forecast a rise at 0 per cent of origins, its mean error was -0.0393 against -0.0055 for the combination, and squared bias is 0.49 per cent of its mean squared error. The difference in squared bias is 55 per cent of the gap in mean squared error between the AR(1) and the combination, and the rest is variance. A mean-reverting model does better in a sample where the series reverts. We do not claim that the downward lean is the full explanation or that it persists.

Gradient boosting scored 0.5745, 2.2 per cent above the AR(1) and 2.6 per cent above the random walk. Neither difference is statistically significant (p = 0.363 and 0.273). The M5 accuracy results paper (Makridakis,

Spiliotis and Assimakopoulos, 2022, International Journal of Forecasting) reports that the top methods were machine-learning methods, mostly LightGBM, and that combinations helped. So that evidence does not say that simple methods beat flexible ones, and our boosting result covers one specification on one series. It is also sensitive to the release-lag assumptions: with every non-spot predictor a week later its RMSE moves from 0.5745 to 0.5638, while the combination moves from 0.5599 to 0.5598.

The AR(1) and the combination have MAEs of 0.4478 and 0.4495. Over 210 rolling 52-week windows the combination had the lower RMSE in 67.6 per cent of them, with a ratio to the AR(1) between 0.9800 and 1.0234. Adjacent windows share most of their origins, so that is one comparison seen through a moving frame.

# 7. Limitations, exploratory status and a prospective test

All 2021 to 2025 results are exploratory. The window was scored repeatedly during development and repair, benchmarks and ablations were added after results had been seen, and no new holdout was added. The two prespecified comparisons are labelled in the code and the outputs, but nothing registered them outside the repository. Neither prespecified comparison rejects at 5 per cent, and none of the 21 exploratory comparisons survives the Holm adjustment (smallest adjusted p = 0.201).

Tables 7.1 and 7.2 give the number of weekly origins at which a two-sided 5 per cent test would detect an RMSE reduction fixed in advance, with 80 per cent power. They depend on their assumptions: the standard deviation of the loss differential and the benchmark's MSE stay at their 2021 to 2025 values, and the effect sizes are round numbers, not a claim about what has been achieved elsewhere. Against the AR(1), a 2 per cent reduction needs 172 origins (3.3 years) and a 1 per cent reduction 680 (13.1 years). Against the random walk the requirements are small only because the combination and the random walk are nearly the same forecast, which keeps the loss differential's standard deviation at 0.0243. A 1 to 5 per cent reduction between two forecasts that differ by 0.089 per cent of spot is implausible, so those rows show what the test could detect, not what a holdout should be expected to find. The table says nothing about any other forecaster.

| RMSE cut | Origins needed | Years of weekly origins | Power with 261 origins |
|---|---|---|---|
| 1% | 680 | 13.1 | 41% |
| 2% | 172 | 3.3 | 93% |
| 5% | 29 | 0.5 | 100% |

Table 7.1. Planning table against the AR(1).

| RMSE cut | Origins needed | Years of weekly origins | Power with 261 origins |
|---|---|---|---|
| 1% | 120 | 2.3 | 99% |
| 2% | 31 | 0.6 | 100% |
| 5% | 5 | 0.1 | 100% |

Table 7.2. Planning table against the random walk.

A prospective holdout would run in five steps. (1) Freeze the design: record the git commit, the chosen configuration and the raw-data snapshot date, publish them before the first new origin, and change nothing afterwards except recorded bug fixes. (2) Prespecify the two comparisons, squared error, Newey-West lag 3, two-sided 5 per cent, and treat everything else as exploratory with a Holm adjustment. (3) Fix the sample size and stop date from the planning table before the first origin. (4) Score each origin once, after its target is fixed, in a log that is only appended to. (5) Report the result whatever it is.

Overlapping four-week forecasts and the overlapping cuts (the strict cut, by-year results and rolling windows) are views of the same 261 errors, not independent checks. The release lags are partly judgement. We tested one alternative, a further week for every non-spot predictor: the combination's RMSE moved by 0.00033 or less for every model except gradient boosting, which moved by 0.0107. We make no claim about profitability. RMSE

and trading return are different objectives.

The first extension we would try is option-implied volatility and risk reversals, since the SARB's working paper WP/22/14 reports that information about the distribution of rand returns improves its forecasts. After that: Bayesian model averaging over a wider component set, horizons of 1, 2, 8 and 12 weeks, and density rather than point forecasts, which matter more to anyone deciding whether to hedge.

# 8. Code and reproducibility

The full pipeline reruns end to end with one command, python src/run_all.py. By default it needs no network access, no paid data and no API key, because the raw files are committed. Every number in this report is read from the saved JSON outputs at build time, so the text cannot drift from the results. All 261 forecasts are logged to forecast_log.csv with their origin date, target date, each model's forecast and the realised value, so each RMSE here can be recomputed from that file in a single line.

# 9. References


- Andrews, D. W. K. (1991). Heteroskedasticity and autocorrelation consistent covariance matrix estimation. Econometrica 59(3).

- Diebold, F. X. and R. S. Mariano (1995). Comparing predictive accuracy. Journal of Business and Economic Statistics 13(3), 253-263.

- Holm, S. (1979). A simple sequentially rejective multiple test procedure. Scandinavian Journal of Statistics 6(2).

- Makridakis, S., E. Spiliotis and V. Assimakopoulos (2022). M5 accuracy competition: results, findings, and conclusions. International Journal of Forecasting.

- Meese, R. and K. Rogoff (1983). Empirical exchange rate models of the seventies: do they fit out of sample? Journal of International Economics 14, 3-24.

- Newey, W. K. and K. D. West (1987). A simple, positive semi-definite, heteroskedasticity and autocorrelation consistent covariance matrix. Econometrica 55(3).

- Rossi, B. (2013). Exchange rate predictability. Journal of Economic Literature 51(4), 1063-1119.

- South African Reserve Bank (2022). Working Paper WP/22/14: can the distribution of rand returns improve exchange rate forecasts?